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Journal of Chemical Theory and Computation
American Chemical Society

Dear Editor,

We ask you to consider the enclosed manuscript, "Basin hopping on Lennard-Jones clusters with one preset at every size," as an Article.

This submission replaces the preprint arXiv:2302.02811, deposited in 2023 under the title "Unified Software Design Patterns for Simulated Annealing." The title has changed. The manuscript now reports one basin-hopping preset. That preset keeps the move library, the temperature, the census, and the orbit completion fixed at every size. Below 98 atoms it adds a diameter penalty, and restart on stall stays off. The novelty is that fixed core, with the depth posterior choosing the move. On the water hexamer, the six crest rows are single draws from the conformer-rotamer ensemble sampling tool (CREST). Those rows are not a success rate. Five polished gaps lie within 1×10^{-7} Eh, and seed 1 lies above that line. The comparison sets the 4000-call arm against CREST energy and gradient calls, which run from 142110 to 209931.

One constructor states the size switch and the stall rule that the code runs. A referee can check each hexamer gap against the force criterion and against the vtight threshold in the text. Journal of Chemical Theory and Computation publishes software for molecular simulation. This article belongs in that journal. The molecular test is a water hexamer at the GFN2-xTB model, and the software is the preset a reader can rerun.

A table of contents graphic at 3.25 in by 1.75 in accompanies the submission.

Suggested reviewers follow.

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Sincerely,
Rohit Goswami